

K—

$$J_x \ddot{\varphi}(t) = K u(t) + M_w(t).$$

2.3 状态空间模型

定义 2.3.1 状态空间模型:

$$\dot{x}_1(t) = \varphi(t), \quad \dot{x}_2(t) = \dot{\varphi}(t).$$

定义 2.3.2 状态空间模型:

$$\begin{aligned} \dot{x}_1(t) &= x_2(t), \\ \dot{x}_2(t) &= \frac{K}{J_x} u(t) + \frac{1}{J_x} M_w(t). \end{aligned}$$

定义 2.3.3 状态空间模型:

$$y(t) = x_1(t) + v(t).$$

定义 2.3.4 状态空间模型:

$$\dot{x}(t) = A x(t) + B u(t) + G w(t), \quad y(t) = C x(t) + v(t),$$

其中:

$$A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ K/J_x \end{bmatrix}, \quad G = \begin{bmatrix} 0 \\ 1/J_x \end{bmatrix}, \quad C = \begin{bmatrix} 1 & 0 \end{bmatrix}.$$

该模型描述了系统的动态特性，其中 $x(t)$ 是状态向量， $u(t)$ 是输入， $w(t)$ 是扰动， $y(t)$ 是输出， $v(t)$ 是测量噪声。

2.4 传递函数模型

定义 2.4.1 传递函数模型:

$$W(s) = \frac{\Phi(s)}{U(s)} = \frac{K}{J_x s^2}.$$

该模型描述了系统的频率特性，其中 $\Phi(s)$ 是输出信号的拉普拉斯变换， $U(s)$ 是输入信号的拉普拉斯变换。

3 系统辨识

3.1 最小二乘法

最小二乘法是一种常用的系统辨识方法，它通过最小化观测数据与模型预测之间的误差平方和来估计模型参数。

3.2 卡尔曼滤波

卡尔曼滤波是一种递归的估计方法，它利用 IMU 提供的加速度和角速度数据来估计系统的状态。

3.3 神经网络

Figure 4.1: Block diagram of a PID controller. The error signal $e(t)$ is the input to the PID controller, which outputs the control signal $u(t)$. The control signal $u(t)$ is the input to the plant, which outputs the system response $y(t)$. The system response $y(t)$ is compared with the reference signal $r(t)$ to produce the error signal $e(t)$.

4 PID Controller

4.1 PID Controller

The PID controller is a control loop feedback mechanism (see Figure 4.1) that calculates the error between the reference signal and the system response, and then generates a control signal to drive the system response towards the reference signal.

- The error signal $e(t)$ is the input to the PID controller.
- The control signal $u(t)$ is the output of the PID controller.
- The system response $y(t)$ is the output of the plant.

4.2 PID Controller

$$u(t) = K_p e(t) + K_i \int_0^t e(\tau) d\tau + K_d \frac{d e(t)}{dt}$$

where K_p , K_i , and K_d are the proportional, integral, and derivative gains, respectively.

$$e(t) = r(t) - y(t)$$

4.3 PID Controller

The PID controller is a control loop feedback mechanism (see Figure 4.1) that calculates the error between the reference signal and the system response, and then generates a control signal to drive the system response towards the reference signal.

$$F(s) = \frac{(K_p + K_i/s + K_d s)}{K/(J_x s^2)} \{1 + (K_p + K_i/s + K_d s) / K/(J_x s^2)\}$$

The PID controller is a control loop feedback mechanism (see Figure 4.1) that calculates the error between the reference signal and the system response, and then generates a control signal to drive the system response towards the reference signal. The PID controller is a control loop feedback mechanism (see Figure 4.1) that calculates the error between the reference signal and the system response, and then generates a control signal to drive the system response towards the reference signal.

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5 Simulation

5.1 Simulation

The simulation is implemented in Python using the `src/simulation.py` file. The simulation is implemented in Python using the `src/simulation.py` file.

- The reference signal $r(t)$ is the input to the simulation.
- The PID controller is implemented using the `PID` class.
- The system response $y(t)$ is the output of the simulation.
- The control signal $u(t)$ is the output of the PID controller.


```

1 ##### PID-#####
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1 ##### `src/simulation.py`
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#####.

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- ϕ — #####;
- $\phi_{\square}$ — #####;
- $\phi_{\square}$ — #####;
- $\phi_{\square\square}$ — #####;
- u — #####;
- M — #####;
- M_w — #####;
- y — #####;
- v — #####;
- J_x — #####;
- K — #####;
- K_p, K_i, K_d — ##### PID-#####.

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